

LUNG CANCER SURVIVAL ANALYSIS

Final Capstone Project



1) OBJECTIVE

Analyze patterns related to lung cancer survival and evaluate predictive models using a synthetic healthcare dataset.



2) DATASET

Kaggle Lung Cancer Dataset (synthetic)



3) TOOLS



Google BigQuery



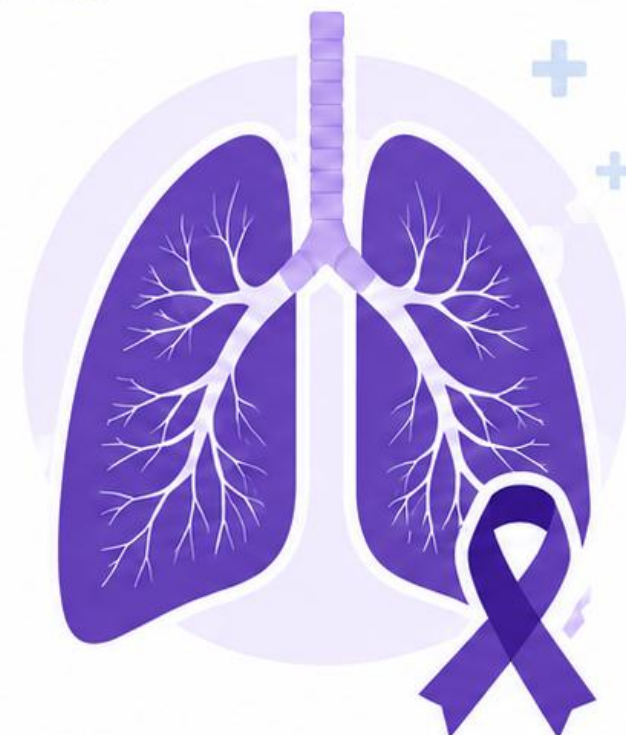
Google Colab



Power BI



GitHub



Source: Kaggle



Workflow: BigQuery → Colab → Power BI → GitHub



Write by: Angela Rodriguez

DATA & METHODOLOGY

Dataset summary and analytical workflow



DATASET SUMMARY



Synthetic Kaggle dataset

Lung Cancer Dataset (synthetic)



890K rows

High-volume synthetic records



17 columns

Clinical, demographic & lifestyle features



Loaded into BigQuery

Data stored and queried in the cloud



Data downloaded as CSV from Kaggle and used for educational and analytical purposes only.



ANALYTICAL WORKFLOW

1



Data collection

Downloaded synthetic Kaggle dataset (CSV).

2



BigQuery loading

CSV file ingested and loaded into BigQuery.

3



Data cleaning

Handled missing values, removed duplicates, and standardized data.

4



Exploratory Data Analysis (EDA)

Explored distributions, relationships, and key insights.

5



Feature engineering

Created new features and encoded categorical variables.

6



Predictive modeling

Built and evaluated models to predict lung cancer survival.

7



Results analysis & conclusions

Interpreted findings and summarized key conclusions.

8



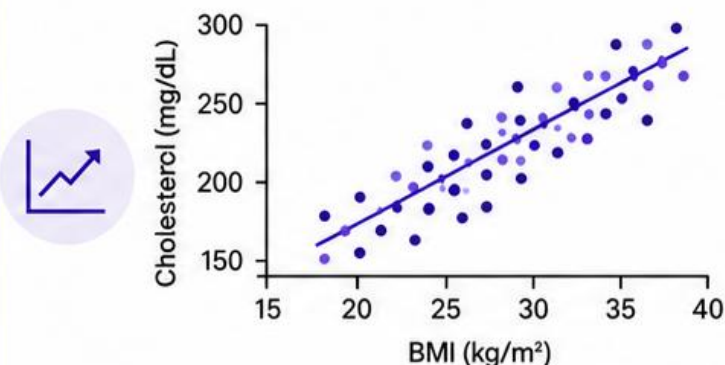
Power BI dashboarding and reporting

Interactive dashboards and reports for insights and decision-making.

KEY PATTERNS / FINDINGS

Main insights from exploratory analysis

1 BMI-cholesterol correlation = 0.75



Interpretation

Higher BMI is associated with higher cholesterol levels in the synthetic dataset.

2 Survival imbalance

22.02%
Survived

77.98%
Did not survive



Interpretation

The dataset is imbalanced, with significantly more non-survivors. Models should account for this to avoid bias.

3 Balanced distributions



Cancer
Stages



Smoking
Status



Family
History

Cancer stages, smoking status, and family history are mostly balanced in the synthetic data.

Interpretation

No major class skew detected for key categorical features.

MODEL RESULTS & CONCLUSION

Logistic Regression vs Random Forest



Logistic Regression

Accuracy	0.49
ROC-AUC	0.50
Precision (Survived)	0.22
Recall (Survived)	0.52
F1-Score (Survived)	0.31



Random Forest

Accuracy	0.55
ROC-AUC	0.50
Precision (Survived)	0.22
Recall (Survived)	0.42
F1-Score (Survived)	0.29

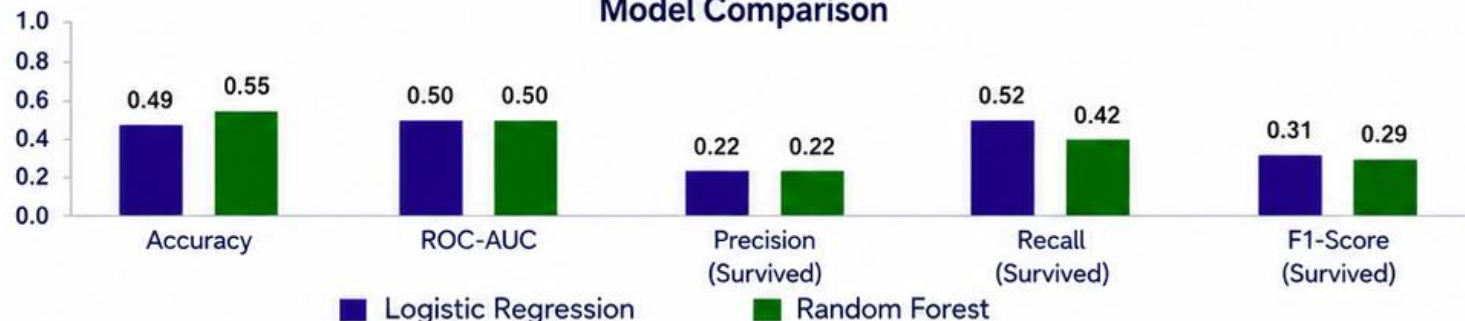


CONCLUSION

- ✓ Random Forest slightly improved accuracy
- ✓ Both models have ROC-AUC near 0.50
- ✓ Predictive power is low due to the synthetic and imbalanced data
- ✓ Future work: test with real clinical datasets and improved feature engineering



Model Comparison



PATTERN ANALYSIS QUESTIONS & FINDINGS

Clear answers to the exploratory research questions

1



Distribution of variables

- ✓ Dataset has ~**890,000** records and **17** features.
- ✓ Numerical variables (age, BMI, cholesterol) show **broad distributions**.
- ✓ Categorical variables (gender, family history, cancer stage, smoking status) are **mostly balanced**.
- ✓ This suggests a **synthetic structure**.

2



BMI and cholesterol

- ✓ This is the clearest relationship in the analysis.
- ✓ Correlation between BMI and cholesterol is about **0.75**.
- ✓ Higher BMI is associated with **higher cholesterol levels**.

3



Survival by stage and treatment

- ✓ Survival rates are very similar across treatments and cancer stages.
- ✓ Surgery has the highest survival at **22.15%**.
- ✓ Chemotherapy has the lowest survival at **21.87%**.
- ✓ Stage survival ranges from **21.81% to 22.14%**.
- ✓ Takeaway: **no strong association**.

4



Smoking status and family history

- ✓ Smoking status is distributed almost evenly across stages, around **25%** in each stage.
- ✓ Family history shows nearly identical survival rates.
- ✓ Takeaway: **no strong visible pattern**.

5



Pre-existing health conditions

- ✓ Hypertension, asthma, cirrhosis, other cancer, and comorbidity status show only **minimal differences** in survival.
- ✓ Takeaway: **no strong association** with survival.



MAIN TAKEAWAY:

Most categorical variables look balanced, while BMI and cholesterol show the strongest visible relationship.



Source: BigQuery



Tools: Colab and Power BI



Write by: Angela Rodriguez

PREDICTIVE MODELING QUESTIONS & FINDINGS

Simple answers from the modeling stage

1



Can the variables predict survival?

- ✓ Not strongly.
- ✓ Logistic Regression achieved accuracy 0.49 and ROC-AUC 0.5029.
- ✓ Random Forest achieved accuracy 0.55 and ROC-AUC 0.5010.
- ✓ Both models are close to random guessing.

2



Which variables contributed most?

- ✓ Random Forest highlighted BMI, age, cholesterol level, family history, gender, cirrhosis, asthma, hypertension, and other cancer as the most relevant predictors.
- ✓ Interpret with caution because the overall model performance is weak.

3



How did class imbalance affect results?

- ✓ The target distribution is imbalanced:
Did Not Survive  77.98%
Survived  22.02%
- ✓ Accuracy alone can be misleading, so precision, recall, F1-score, confusion matrix, and ROC-AUC were used.

4



How well did the models perform beyond accuracy?

Metric	Logistic Regression	Random Forest
 Precision (Survived)	0.22	0.22
 Recall (Survived)	0.52	0.42
 F1-Score (Survived)	0.31	0.29

- ✓ Both models struggled to correctly identify survivors.

5



Did feature engineering help?

- ✓ Created gender_encoded, family_history_encoded, bmi_group, cholesterol_group, and has_comorbidity.
- ✓ These helped organize and interpret the data, but they did not substantially improve predictive power.



CONCLUSION

Overall conclusion: the synthetic dataset contains limited predictive signal for survival, and both models show low predictive power.



Source: BigQuery



Tools: Colab and Power BI



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MAIN CONCLUSIONS

Key takeaways from exploratory analysis and modeling

1 Strongest Pattern



BMI and cholesterol level showed the clearest relationship in the dataset, with correlation ≈ 0.75 .

Patients with higher BMI generally had higher cholesterol levels.

2 Survival Differences Were Small



Most categorical and cross-tabulation analyses showed weak or minimal differences in survival across treatment type, cancer stage, family history, smoking status, hypertension, asthma, and cirrhosis.

3 Imbalanced Target



22.02%
Survived

77.98%
Did not survive



The target variable was imbalanced: 77.98% did not survive and 22.02% survived.

4 Model Performance Was Weak



Logistic Regression

✓ Accuracy: **0.49** | ROC-AUC: **0.5029**

Random Forest

✓ Accuracy: **0.55** | ROC-AUC: **0.5010**

Random Forest was slightly better, but neither model showed strong predictive power for survival.



Overall, the project found clear descriptive patterns, but the selected variables did not provide strong predictive power for survival outcome.

RECOMMENDATIONS & NEXT STEPS

Improving future analysis and clinical relevance

1 Educational Use



This project is useful for exploratory and educational purposes, but it should not be used for real clinical prediction because the dataset is synthetic.

2 Use Real-World Data



Future studies should use real clinical or public health datasets from sources such as CDC, SEER, state cancer registries, hospitals, and healthcare organizations.

3 Add Stronger Predictors



Include more clinically relevant variables such as tumor size, cancer subtype, treatment duration, treatment response, medication history, genetic markers, laboratory results, smoking intensity, time since diagnosis, and follow-up period.

4 Improve Modeling



Continue feature engineering and test better predictive approaches on real data. Current models lacked reliable predictive power, but the workflow provides a strong foundation.



Final takeaway: The workflow from BigQuery to Colab to Power BI was effective for analysis and reporting, but stronger data and more informative features are needed to build reliable survival prediction models.



THANK YOU

Lung Cancer Survival Analysis

